Welcome

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Objectives for the course

- A broad introduction to energy technologies today and in the future
- Most sessions will present one of the main energy technologies and try to put it in a future perspective.
- A few sessions are overarching and treat general physical principles related to energy and energy economics.
- The form is about two hours of lecturing followed by two hours of exercises (problems)

Lecture plan

#	Title	Content	Teacher
		Introduction	
1 (1/9)	Introduction	Welcome and outline of the course. Basic principles for energy.	Oluf
2 (8/9)	Thermodynamics and thermal energy	The need to know thermodynamics for energy technology. Thermal engines, heat pumps and geothermal energy.	Oluf
		Energy sources	
3 (15/9)	Solar	Solar resource, solar thermal energy, and photovoltaics.	Rafael
4 (22/9)	Wind	Wind resource and wind turbines (other mechanical systems in brief)	Rafael
5 (29/9)	Biomass	Biomass resource, thermochemical conversion, and biofuels. (other bio-e-fuel concepts in brief)	Rafael
6 (6/10)	Nuclear	Fission and fusion.	Oluf
		Autumn holiday	
		Energy conversion, transmission and storage	
7 (20/10)	Electrochemical cells	Basics of electrochemisty and electrochemical cells.	Oluf
8 (27/10)	Hydrogen and Power-to-X	Fuel cells, electrolyzers and hydrogen as an energy carrier. Electrofuels.	Oluf
9 (3/11)	Infrastructure	Electricity transportation principles and the energy grid (other energy integration infrastructure in brief)	Rafael
10 (10/11)	Energy storage	The different technologies. Choice and scale.	Oluf
11 (17/11)	Batteries	Batteries and flow batteries.	Oluf
		Wrapping up	
12 (24/11)	Energy economics	Basic parameters and trends in energy economics	Rafael
13 (1/12)	Wrap-up	Missing bits and pieces. Wrap-up.	Oluf
15/12	Exam (E4A)	Multiple choice 2 hours (*time of the day pending*)	

Teachers



Professor

Jens Oluf Jensen



Assistant professor

Rafael Nogueira Nakashima

Teaching Assistants (TAs)

Will assist you during problems solving



Waleed Khodery



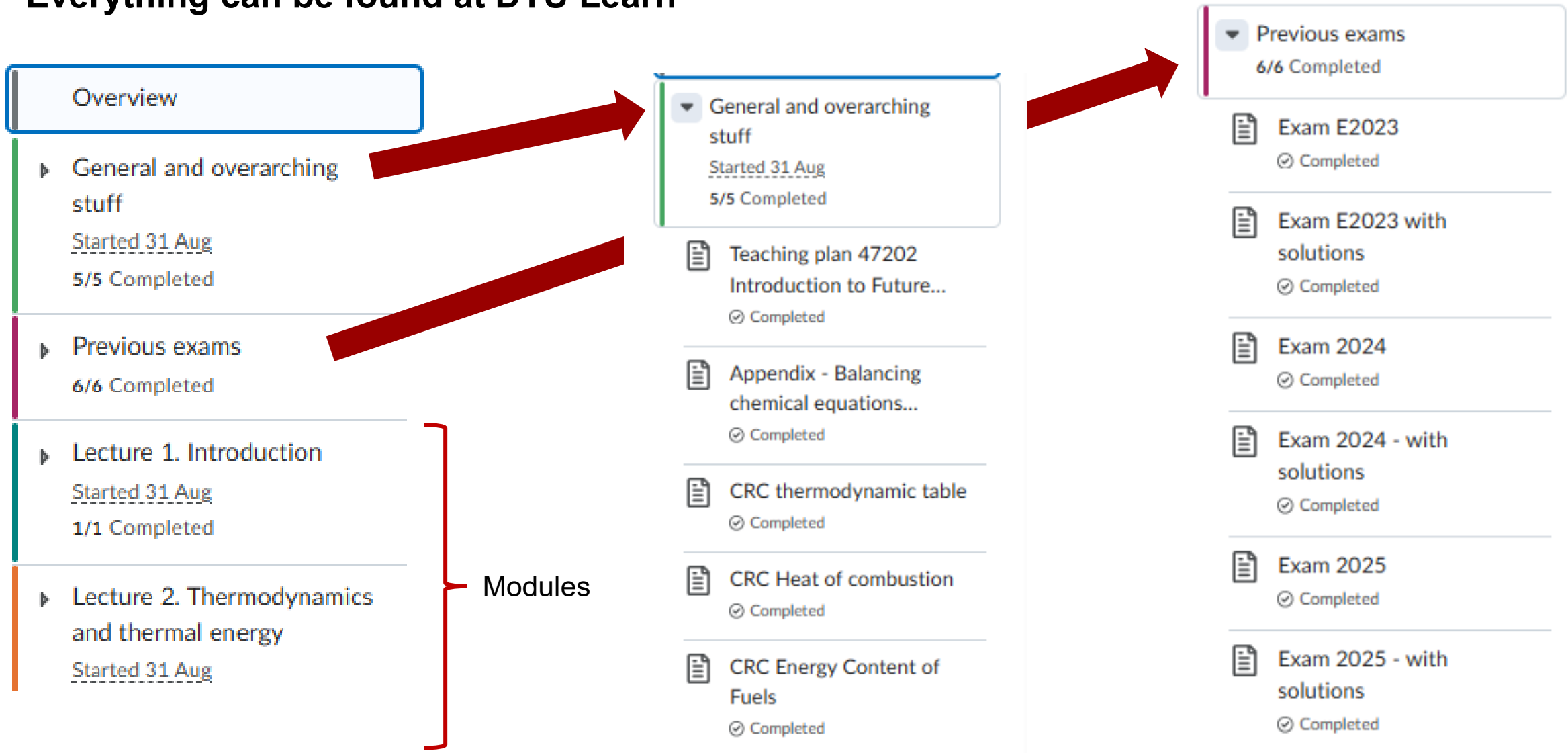
Amirhosein Farmani



Nayan Biswas

The elements of the course

Everything can be found at DTU Learn



The elements of the course

In common modules:

- Lecture plan
- Leaning objectives for each session
- List of formulas
- Previous exams with solutions
- Supplementary literature (optional)
- …

In 13 lecture modules:

- Texts for preparation
- Slides for the lectures
- Exam-like questions (Quizz)
- Problems and solutions
- Supplementary literature (optional)
- …

Lecture modules are updated during the course

The exam – Multiple choice

- Log in to https://eksamen.dtu.dk
- Pin code will be given at location.
- At the exam, you will have 2 questions from each of the first 12 lectures.

  (24 questions in total).
- There will be 5 possible answers for each question, but only 1 correct.
- No negative points for wrong answers
- Some questions will require calculations, others will not.
- All aid allowed (all teaching material, slides, exercise solutions, previous exams …) - but no open internet

Previous exam question available at Learn. Don't get confused.